

“Telecom T&M instruments have a high level of sophistication”



Electronic test and measurement (T&M) instruments are indispensable for validating the functioning and performance of electronic products. P.S. Deodhar, chairman of Aplab, one of India's leading manufacturers and exporters of test & measurement instruments, spoke to EFY's Uma Gupta about the T&M industry, outlining key trends and recent activity

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What are the top tech trends that are changing the test-ing landscape world over?

Confluence and convergence of computers and communications, and the emergence of automation in a wide range of industrial systems (and even consumer products), has not only created a bigger demand for T&M products but also posed new challenges to their technical capabilities. Today, all T&M products have to be intelligent and digitally addressable, both for data acquisition and control. They need to be fully digitally controlled using microprocessors and digital signal processors. Also, they need to be capable of self-calibration and user-friendly data acquisition.

There is a need for T&M equipment that are capable of measuring high-rising pulses, glitches and fast-changing digital signals and working in mixed domain. Broadband signal measurement with modulation and demodulation measurements, and simulation with computer for in-depth analysis is in high demand.

At the same time, these high-performance equipment are needed in a portable form. Complex digital modulation, e.g., orthogonal frequency-division multiplexing, and measurements on wide bandwidths have become commonplace.

In India, what are the top three markets for test and measurement that have come up recently?

In India, telecom is today the largest market for T&M products as telecom has grown both terrestrially and via satellites. The sophistication here is high and it covers a wide range of techniques and measurement topologies. The next large market is education and training. Engineering education is growing and so is the market. Fast obsolescence is another reason for market growth. T&M products are also needed in applications such as defence and industrial automation.

In India, there is very little hardware manufacturing being done both in telecom and IT fields. Not even 5 per cent of the hardware needed for these two sectors is produced in India. It is a direct result of the lack of planning by the government as well as the industry. As a consequence, T&M demand for manufacturing, like for calibration and quality assurance, does not exist.

Demand is primarily for maintenance of the user equipment. Here again fast technology changes lead to products getting obsolete before their useful life is over. There is no great need for repair and reuse like 25 years ago. We have therefore ended up with huge electronic garbage to deal with.